

INF2004 Embedded Systems Programming

Module Overview

Who are we?

Lecture

- Muhamed **Fauzi** Bin Abbas
  (Module Lead)
  - *fauzi.abbas@singaporetech.edu.sg*

Lab Manager

- **Haja** Mohideen S/O Packir Rowther
  (Senior Assistant Manager)
  - *haja.rowther@singaporetech.edu.sg*



Email header: **[INF2004] xxxxxxx**

compulsory

Lab Instructors



- **Aw** Kok Seng
  (Lead Professional Officer)
  - *kokseng.aw@singaporetech.edu.sg*
- **Deric** Ng
  (Senior Professional Officer)
  - *deric.ng@singaporetech.edu.sg*
- **Paul** Fong
  (Adjunct Faculty)
  - *af_lee.kim.fong@singaporetech.edu.sg*
- **Kang** Liat Chuan
  (Adjunct Faculty)
  - *af_liat.chuan.kang@singaporetech.edu.sg*

Teaching Activities

- **Lectures** - **2** hours per week (via Zoom)
  - Monday, 11.00 AM – 1.00 PM *** Week 11: No Lecture*

- **Laboratory** – **2** hours per week starting from **Week 1**

Lab Group	Day	Timeslot	Location	Lab Instructors
P5 (IS-DCO)	Monday	4 pm - 6 pm	E2-05-14	Fauzi, Kang
P6 (IS-DCO)	Monday	4 pm - 6 pm	E2-05-05	Aw, Fong
P2 (AAI)	Tuesday	2 pm - 4 pm	E2-05-07	Aw, Kang
P3 (CSC)	Wednesday	11 am – 1 pm	E2-05-14	Fauzi, Deric
P4 (CSC)	Wednesday	11 am – 1 pm	E2-05-05	Aw, Kang
P1 (SE)	Thursday	4 pm – 6 pm	E2-05-14	Fauzi, Paul

How are the Lectures conducted?

TLDR; We flip you then we guide you. ☺

How are the Lab sessions conducted?

We guide you... Then we test you. :P

How are the Project work conducted?

Own Time, Own Target for six weeks with check-ins.

How are you Assessed?

- **Lecture Quiz (30%)**
  - Test your understanding of the lecture concepts.
  - xSITe-based quiz, i.e. mcq, fill in the blanks, etc
  - Conducted in **Week 12**.
  - Each week, a topic quiz will be released, and you can attempt it unlimited times, and it is not graded. It will randomly pick from a set of 5 questions.
  - In Week 12, you will have ONLY 1 attempt to sit for the quiz that will comprise **all** topics and be graded.
    50 Questions in 30 minutes!
  - There will be a notification on xSITe if the weekly question pool is updated.

How are you Assessed?

- **Lab Assignments** (**10%** = 2% x 5)
  - Test your knowledge based on your lab sessions.
  - Submit a mini code assignment for each lab session.
    - due 2 weeks after the actual lab.
  - Lab for Week 2 to Week 6
- **Practical Skills Assessment** (**15%**)
  - 30 mins individual in-person assessment in Week 12 (TBC)

- **Group Project** (**45%**)
  - Test your ability to work on a complex project
  - Team comprised of 4-5 students
  - Team members must be within the same timeslot sessions
  - Peer Evaluation
    - Project will be assessed holistically
    - Individual team members scores are moderated based on their contributions (via various method mostly peer evaluation)
  - Five grading stages:
    - [W6] Requirements + Design (Hurdle)
    - [W8] Tribunal 1 – 15
    - [W10] Subsystems Check-in (Hurdle)
    - [W13] Tribunal 2 - 15
    - [W13] Final Run (Demo + Final Code + Testing) - 15

** **Hurdle** = If you miss or don't submit on time, 5% off total group project marks every day.

Assessment (in Summary)

Activities	%	Deadline	Assessment Type
Lab Assignments	10%	A week after each lab	Submit a mini code assignment
Design Package Deliverables	Hurdle	Week 6 (Sat, 2359 hrs)	Submit a 3-5 pages document
Tribunal 1	15%	Week 8 (Lecture slot)	Zoom Discussion
Lecture Quiz	30%	Week 12 (Lecture slot)	MCQ quiz
Practical Skills Assessment	15%	Week 12 (Lecture slot)	Individual in-person assessment
Tribunal 2	15%	Week 13 (Lecture slot)	Zoom Discussion
Final Demo (Project)	15%	Week 13 (Lab Sessions)	Final Demo + Code/Test Review

Additional Reference Materials

- Embedded C Coding Standard (https://barrgroup.com/sites/default/files/barr_c_coding_standard_2018.pdf)
- Michael Barr, Programming embedded systems in C and C++ (https://barrgroup.com/programming-embedded-systems-c-and-c)
- C-programming (https://www.cprogramming.com/tutorial/c-tutorial.html)
- What is Unary Operator in C? (https://www.scaler.com/topics/unary-operator-in-c/)
- C Programming Language Tutorial. (https://www.geeksforgeeks.org/c-programming-language/)

Reference Materials (Raspberry Pi Pico)

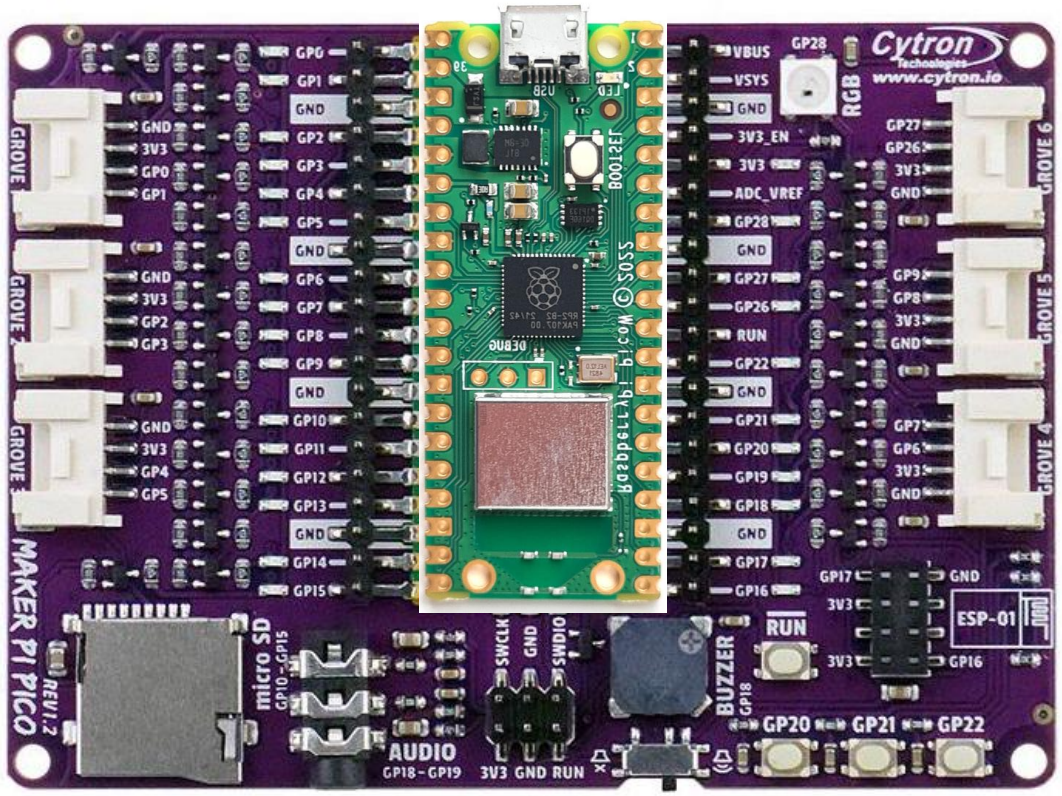
SINGAPORE INSTITUTE OF TECHNOLOGY

1. Official Raspberry Pi Pico Documentation: The official Raspberry Pi Foundation provides extensive documentation on the Pico, covering topics from getting started to detailed hardware specifications. (https://www.raspberrypi.org/documentation/rp2040/getting-started/)
2. Raspberry Pi Pico Datasheet: A detailed technical document outlining the hardware specifications and capabilities of the RP2040, the microcontroller chip on the Pico. (https://datasheets.raspberrypi.org/rp2040/rp2040-datasheet.pdf)
3. Raspberry Pi Pico Python SDK: This is the official documentation for the MicroPython environment on the Raspberry Pi Pico. It covers modules, functions, and other MicroPython-specific features. (https://datasheets.raspberrypi.org/pico/raspberry-pi-pico-python-sdk.pdf)
4. Raspberry Pi Pico C/C++ SDK: For those who prefer to program in C or C++, the Raspberry Pi Foundation provides an SDK for that too. (https://datasheets.raspberrypi.org/pico/raspberry-pi-pico-c-sdk.pdf)
5. Pico Examples: The Raspberry Pi Foundation's GitHub repository contains a plethora of example projects and code snippets to help users get started with the Pico. (https://github.com/raspberrypi/pico-examples)
6. Community Forums: The Raspberry Pi Foundation's forums and other online communities are invaluable resources for troubleshooting, project ideas, and general discussions about the Pico. (https://www.raspberrypi.org/forums/)

Module Schedule

	LECTURE (ONLINE)		LABOTATORY (IN PUNGGOL)		NOTE
	1 HOUR	1 HOUR	1 HOUR	1 HOUR	
W1	MODULE OVERVIEW	INTRODUCTION	ISSUE PICO	SETUP	PROJECT BRIEFING (VIDEO)
W2	INTERRUPTS	GPIO & DIGITAL COMMS	GPIO	DIGITAL COMMMUNICATION	
W3	ADC	TIMER & PWM	INTERRUPTS	TIMER	
W4	EMBEDDED OPERATING SYSTEMS	STATE MACHINE	PWM	ADC	
W5	REQUIREMENTS & DESIGN		SENSOR FUSION	EDGE	
W6	TRACEBILITY & TESTING		OPTIMISATION	DEBUGGING	Design Package Deliverables
W7	BREAK WEEK				
W8	TRIBUNAL 1		OWN TIME OWN TARGET (OTOT)		
W9	SENSOR FUSION		OTOT		
W10	OPTIMISATION		SUBSYSTEM CHECK-IN		
W11	EDGE COMPUTING (PRE-RECORDED VIDEO)**		OTOT		** NO LECTURE
W12	LECTURE QUIZ	PRACTICAL QUIZ	OTOT		
W13	TRIBUNAL 2		FINAL DEMONSTRATION		

Microcontroller board used in this module



Raspberry Pico W + Maker Pi

Ground Rules…

- We are here for those who want to learn
  - Participate in lecture
  - You have a question/doubt, whom better to ask than “yours truly” ☺
  - Speak now or forever hold your peace…
  - **All queries should go to the discussion forum, especially if they are topic-related**.
- This module has no exams (most of you asked for it during your video interview)
- We generally assess you on materials covered in lectures & lab sessions…
- Discussion is encouraged, BUT plagiarism will not be tolerated!
  - Gen AI??
- Please ensure your laptop is working before any assessment!
  - I will not arrange make-up due to laptop/hardware failures.
- If you are unwell, get MC and submit it officially. If you missed any graded assessments, please email the Module Lead (Fauzi) ***until you receive a response***.
- If you have done a reasonably complex Embedded System project before, you may try to submit it as a substitute for your project component. Please drop me an email, and we can discuss further. ☺

any time



assessments



Updates

- Raspberry Pi Pico will only be loaned to you in the Week 1 Lab session.
  - There will be a lab in Week 1 (this week)
- Project components will be entirely issued later (by Week 6).
- Please come for the labs on time (or earlier)
- If you are stuck in your project progress…
  - Ask any of us for assistance.
  - **Keep asking us for help if you are lost/stuck/confused.**
  - Don't be shy; everyone here has paid their school fees…

Email header: **[INF2004] xxxxxxx**

compulsory

Permissions

- Students enrolled in this module automatically grant the Singapore Institute of Technology the use of anonymised work submitted in this module. The usage of submitted work is granted only for educational and research purposes, and the author's identity of any work will not be revealed or implied.
- Students who choose to withhold consent and not give permission for the usage of anonymised submissions are to inform the Module Coordinator via email, before the start of the second session of the module (Week 02).
- Students must seek Module Lead (Fauzi) permission before submitting any work that came from this module for any competition, seminars, workshops, exhibitions, etc.